RETRIEVAL-AUGMENTED GENERATION

How RAG Models Actually Work

A Bee's Guide to Retrieval-Augmented Generation

Presented by Trustan Price

THE SETUP

Why Bees?

I'm going to explain a technical AI concept — Retrieval-Augmented Generation, or RAG — using something familiar: how a beehive finds food.

Every role in the hive maps to a specific part of the system.

Meet the Spelling Family

Three bees, three jobs — no technical terms yet.

Sage Spelling

SCOUT

Flies out first. Searches wide and reports back which patches are worth visiting.

Fern Spelling

FORAGER

Goes wherever Sage points. Collects the real nectar and hauls it home.

Queen Clover Spelling

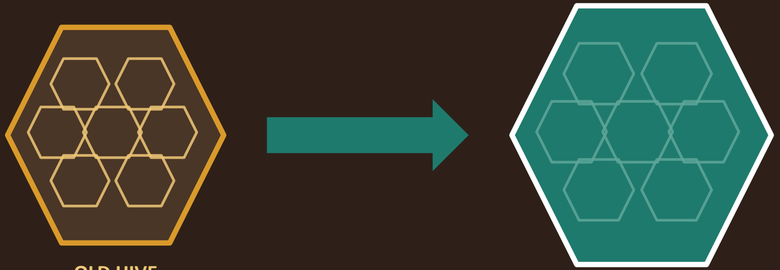
THE HIVE'S DECISION-MAKER

Never leaves the hive. Combines what's brought to her into the final call.

Moving Day

The Spelling family's old hive got knocked down — remodeled into a new bank branch across town. (We won't name names.)

Overnight, the whole colony is relocated to Synchrony Park. Every bee's mental map of “where the good flowers are” is now useless.

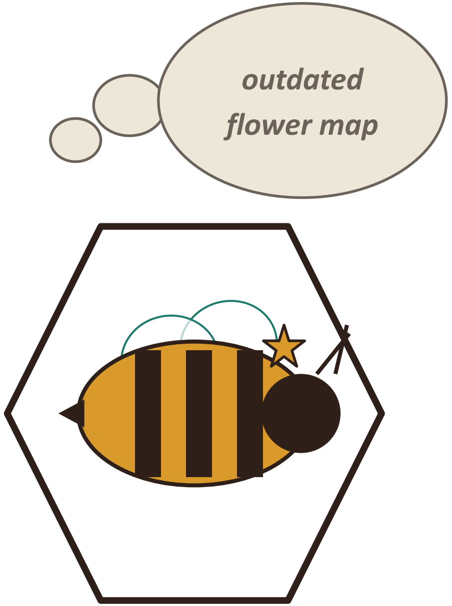


Queen Clover's Problem

Every day, the hive asks Queen Clover the same question: “where should we forage today?”

Her only stored knowledge is a flower map from the old location — memorized perfectly, months out of date, and now geographically useless.

She isn't wrong because she's broken. She's wrong because the world changed underneath her.



THIS IS —

...what happens when an LLM's training data goes stale.

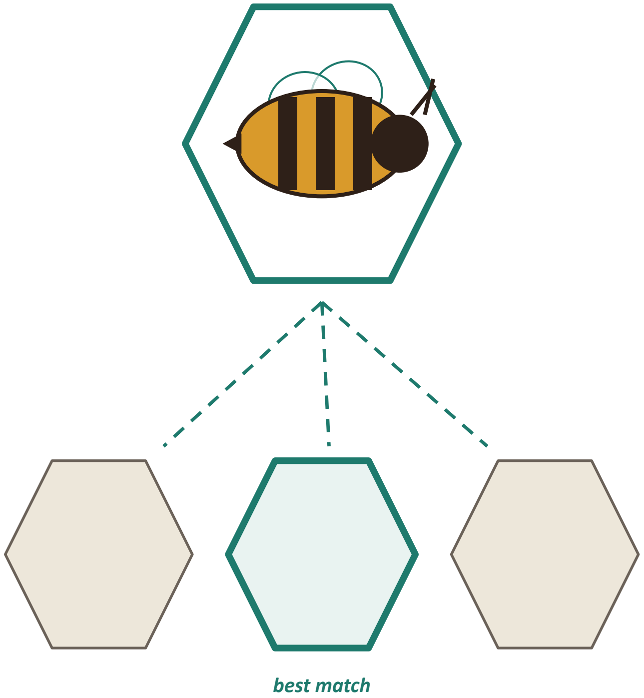
The model isn't broken — it's just answering from memory of a world that has since moved on.

RAG
STEP 0

Sage Takes Flight

Since old memory won't work, Sage flies out into unfamiliar Synchrony Park — checking flower patch after flower patch against what the hive actually needs right now.

She doesn't collect anything herself. She just finds and flags the best matches.



THIS IS —

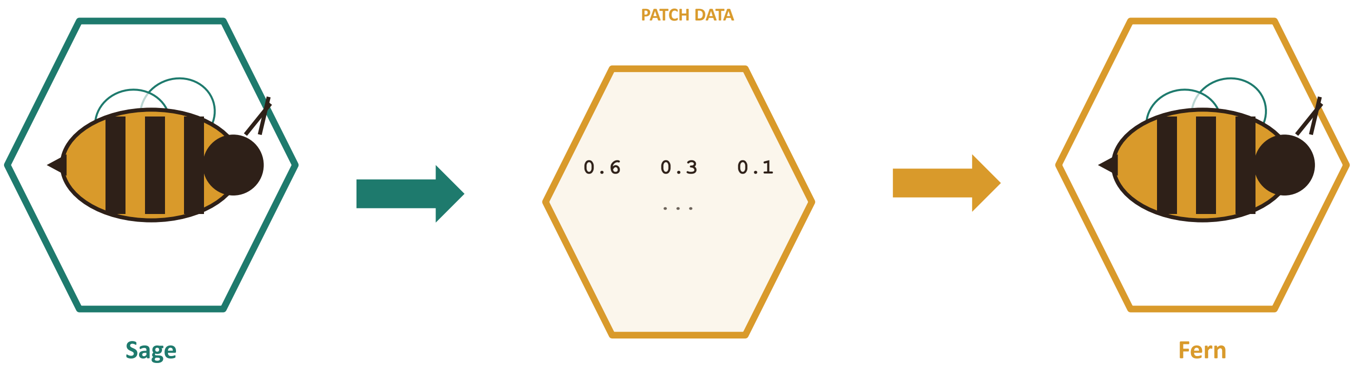
...retrieval. Searching a knowledge base for the most relevant matches to a query.

In a real system, this is a similarity search over vector embeddings — not a keyword match, a meaning match.

RAG
STEP 1

Fern Brings It Home

Sage hands off exactly what she found — not a vague direction, but the real coordinates. Fern flies there, collects the actual nectar, and hauls it back to the hive.



This is the moment a scout hands a forager a vector — not a vague hint, a precise coordinate.

THIS IS —

…augmentation. Turning retrieved information into usable material for the next step.

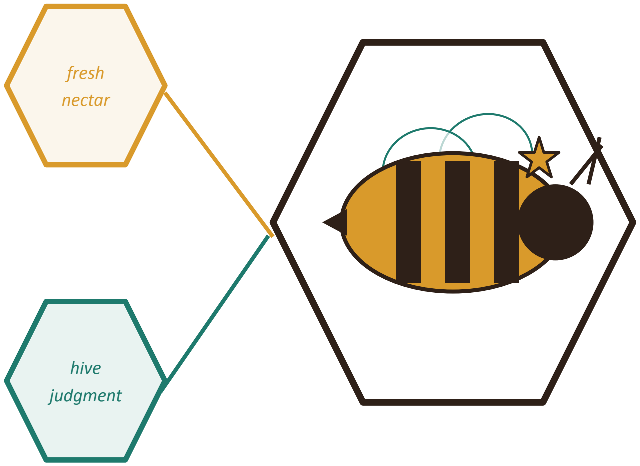
The raw match Sage found becomes real content — inserted directly into the context the Queen will work from.

RAG
STEP 2

Queen Clover Makes the Call

Queen Clover combines what Fern brought back today with her own judgment — how much honey the hive needs, which nectar mixes well — to make the best call.

She's not just repeating what she was handed. She's reasoning over it.



THIS IS —

...generation. The LLM producing an answer grounded in retrieved context, not memory alone.

This is the part people call “the AI” — but it's only as good as what the scouts and foragers brought it.

RAG
STEP 3

Why Not Just Relearn the Park?

RETRAIN THE QUEEN

Every bee re-memorizes the entire park from scratch before answering anything.

- ✕ Slow
- ✕ Expensive
- ✕ Stale again the next time the park changes

RETRIEVE FRESH NECTAR

Don't retrain the Queen — just feed her better information at query time.

- ✓ Fast
- ✓ Cheap
- ✓ Adapts immediately when the park changes

PUTTING IT ALL TOGETHER

The Full Flight Path

Relocation
QUERY

Sage
RETRIEVAL

Fern
AUGMENTATION

Queen Clover
GENERATION

Honey
ANSWER

Query in ⟶ fresh knowledge retrieved ⟶ grounded in context ⟶ answer generated.

A hive that only trusts old memory starves in a new park.

One with active scouts adapts immediately.

That's Retrieval-Augmented Generation.

Thank you.